

PAPER Tandem-Mirror End-Cell and Plug Alternatives for a D–³He Burner: Kinetic-Stabilizer End-Cells, the Mirror-Ratio–End-Loss Optimum, and Expander Flux-Expansion Trades

P I Ford¹ and G L Kulcinski¹ ¹Kronos Fusion Energy, Los Angeles, California, USA

ORCID (P I Ford): 0000-0003-0395-1752

Abstract

The end cell of a tandem-mirror burner must simultaneously keep the plug plasma micro-stable, set the mirror ratio that trades engineering gain against axial end loss, and expand the escaping flux tube onto a survivable end wall—and each of these three tasks admits more than one physical solution. We pose the three problems as a single coupled optimisation for the Aegis deuterium–helium-3 burner, derive the governing relations, and down-select each lever with a GPU-accelerated computation. First, a one-dimensional three-velocity (1D3V) electromagnetic hybrid particle-in-cell (PIC) study of a kinetic-stabilizer end cell shows that an energetic (90 keV) isotropic ion population injected at a stabilizer fraction $f_s \geq 0.30$ drives the plug anisotropy-driven ion-cyclotron microinstability from a normalised growth rate $\gamma/\omega_{ci} = 0.0365$ to 0.0114 and, by $f_s = 0.45$, to full suppression, cutting the saturated field-energy amplification from $145\times$ to $3\times$; a reduced marginal-stability estimate places the threshold at $f_s \approx 0.31$, in agreement with the kinetic result. A cold (3 keV) warm fill of comparable fraction leaves the mode untouched, isolating an energy/velocity-matched resonant-damping mechanism rather than density dilution. Second, a reduced-order power-balance scan over central-cell mirror ratio $R_{mc} = 2.98\text{--}7.07$ (on-axis mirror field 11–26 T) finds the engineering gain rising monotonically, $Q_E = 1.15 \rightarrow 1.55$, against a nearly linear expander-area and plug-field penalty; the burner is fixed at the operating point $R_{mc} = 4.61$ ($B_m = 17$ T), where $Q_E = 1.318$ —the frozen burner gain—behind a plug coil of 26.49 T peak conductor field, a magnet distinct from the breeder’s 16.84 T centrepost. Third, a field-line expander study shows the end-wall heat flux obeys $q_{\text{end}} = (P_{\text{ax}}/A_{\text{thr}}) \Lambda^{-1}$, so bringing the 2.72 GW axial end-loss stream through a 0.752 m^2 throat to the 2.5 MW m^{-2} end-wall limit requires a throat-area flux expansion $\Lambda \approx 1447\times$. The three levers close on one internally consistent end-cell design that delivers $Q_E = 1.318$ with the direct-conversion fraction held at 0.49. The single outstanding refinement is a two-dimensional kinetic confirmation of the stabilizer, which the one-dimensional model cannot supply.

Keywords: tandem mirror; kinetic stabilizer; mirror ratio; Pastukhov end loss; expander; ion-cyclotron microinstability; direct energy conversion; helium-3 burner

Contents

1	Introduction	3
1.1	The end cell governs a tandem mirror	3
1.2	Prior art	3
1.3	Contribution	3
2	The three end-cell jobs and their acceptance metrics	3
3	Governing equations and the formal problem	4
3.1	Ambipolar confinement and the mirror-ratio lever	4
3.2	The anisotropy-driven ion-cyclotron microinstability	4
3.3	The kinetic-stabilizer mechanism	5
3.4	Mirror ratio, engineering gain and the expander penalty	5
3.5	Expander flux conservation and the end-wall constraint	5
4	Computational methodology: the NVIDIA GPU HPC campaign	6
4.1	The 1D3V electromagnetic hybrid particle-in-cell model (AG-CX-03)	6
4.2	Reduced-order mirror-ratio scan and field-line expander integration	6
4.3	Verification, reproducibility and problem scale	7
5	Results	7
5.1	Kinetic-stabilizer suppression	7
5.2	The mirror-ratio optimum	8
5.3	The expander floor	10
5.4	The integrated end-cell design and two distinct magnets	10
6	Discussion	11
6.1	Three coupled optima	11
6.2	Against the published benchmarks	11
6.3	Why the plug potential depends on microstability	12
7	Limitations	12
8	Conclusion	12
	Acknowledgments	12
	Data availability	13
	Competing interests	13

1 Introduction

1.1 The end cell governs a tandem mirror

A tandem mirror confines its central-cell plasma electrostatically, between two high-field plug cells that raise an ambipolar potential peak the central ions cannot climb [9, 10, 11]. Everything that decides whether such a machine gains happens in the end cell. The plug must be magnetohydrodynamically (MHD) and microscopically stable, or the confining potential collapses; the mirror ratio must be high enough to suppress the Pastukhov axial loss [13, 14] but not so high that the plug magnet and the downstream expander become unbuildable; and the flux that does escape must be spread over enough end-wall area to be cooled. These are three coupled engineering choices, each with candidate solutions, and this paper down-selects all three for the Aegis burner around its frozen operating point (engineering gain $Q_E = 1.318$, neutron fraction $f_n = 5.44\%$, direct-conversion fraction 0.49) [1, 2].

1.2 Prior art

The tandem mirror is a mature line. Its confinement principle was demonstrated on TMX and TMX-U and on GAMMA-10 [22], and the thermal-barrier variant of Baldwin and Logan [11] showed how a depressed potential in the plug decouples plug and central-cell electrons to raise the accessible potential. The axisymmetric-mirror revival—driven by high-temperature-superconductor coils—has restored interest in simple, MHD-stable mirror ends: the gas-dynamic trap (GDT) line reached record mirror parameters and electron temperatures [20, 21], the Wisconsin HTS Axisymmetric Mirror (WHAM) established a modern physics basis for a high-field axisymmetric plug [19], and Fowler, Moir and Simonen argued that a kinetic stabilizer can replace the geometric complexity of earlier stable-mirror ends with a much simpler axisymmetric machine [17, 18]. The microstability of anisotropic mirror plasmas—the loss-cone-family drift-cyclotron-loss-cone (DCLC) and the anisotropy-driven Alfvén ion-cyclotron (AIC) branches—has been understood since the finite-length analyses of Post and Rosenbluth [16] and was measured directly on 2XIIB, where a warm plasma stream stabilised the loss-cone mode [23] and an anisotropy-driven ion-cyclotron mode was identified [24]. Direct recovery of the escaping end-loss power, rather than its thermal conversion, has an equally long pedigree in the venetian-blind and plasma direct converters of Moir and Barr [25, 26]. Advanced-fuel ($D-^3\text{He}$) operation, which the burner adopts to suppress neutrons and enable direct conversion, sets stringent confinement requirements that make the end-cell loss budget decisive [27, 28].

1.3 Contribution

Against that background, the contribution here is a single, internally consistent end-cell design point for the Aegis burner, in which each of the three levers is a tested computation rather than an assumption imported from a neighbour. We (i) state the governing relations for the ambipolar confinement, the anisotropy-driven ion-cyclotron microinstability, the mirror-ratio-gain trade, and the expander flux conservation (§3); (ii) describe the GPU-accelerated hybrid-PIC and reduced-order computations that produced the numbers, with their discretisation, solver and verification (§4); (iii) report the down-select of the kinetic stabilizer, the mirror-ratio optimum, and the expander floor (§5); and (iv) show that the three studies close on one design that delivers the frozen burner gain while preserving the direct-conversion fraction. Where the plug coil (26.49 T peak conductor field) and the breeder centrepost (16.84 T) appear together, they are kept explicitly distinct—two different magnets in two different machines. Throughout we cross-reference the companion physics of the burner’s plug potential, its microstability, its magnet and its direct converter [2, 6, 7, 4, 3, 5, 8].

2 The three end-cell jobs and their acceptance metrics

Figure 1 lays out the end cell: the central cell terminates in a mirror throat and plug, an energetic isotropic population is injected to hold the plug micro-stable, and the surviving flux tube fans out through the expander onto the end wall or direct converter. The plug parameters used to score the three jobs are those of the burner plug: on-axis plug field $B_0 = 10$ T in the stability study, ion parallel temperature $T_{i\parallel} = 90$ keV, temperature anisotropy $A \equiv T_{\perp}/T_{\parallel} = 2$, and plug density $n = 4 \times 10^{21} \text{ m}^{-3}$ (a factor ≈ 16 above the central-cell density, the ratio that sets the ambipolar barrier). The plug *coil* peak conductor field is 26.49 T—the governing magnet number, engineered separately in the companion structural study [4], and a different device from the breeder’s 16.84 T centrepost [1]. Table 1 collects the three levers, the tested range, the selected point and the de-risking serial.

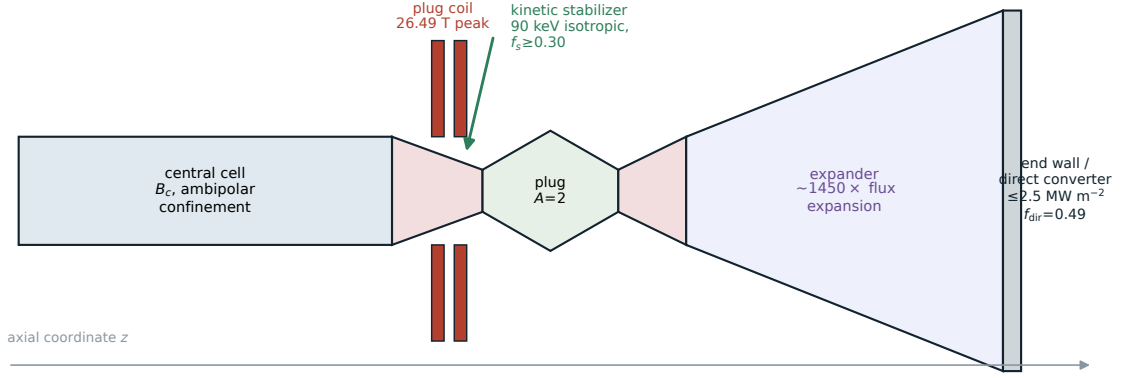


Figure 1. (Schematic.) End-cell layout of the Aegis burner. The central cell (field B_c , ambipolar confinement) terminates in a mirror throat and plug ($A = 2$) behind a plug coil of 26.49 T peak conductor field; an energetic 90 keV isotropic kinetic stabilizer at $f_s \geq 0.30$ holds the plug micro-stable; the surviving flux tube fans out through a $\sim 1450\times$ expander onto the end wall / direct converter at the 2.5 MW m^{-2} limit with direct-conversion fraction 0.49. No data values are plotted; the figure encodes structure only.

Table 1. End-cell and plug design levers for the Aegis burner, with the tested range and the selected point. Serials are Physics De-Risking Register entries [1].

lever	tested range	selected point	serial
kinetic-stabilizer fraction f_s	0–0.45	γ/ω_{ci} suppressed at $f_s \geq 0.30$	AG-CX-03
mirror ratio R_{mc}	2.98–7.07 (B_m 11–26 T)	knee at 4.61 (B_m 17 T), $Q_E = 1.318$	AG-CX-06
expander flux expansion Λ	50–2500×	$\approx 1447\times$ to 2.5 MW m^{-2}	AG-CX-11

3 Governing equations and the formal problem

3.1 Ambipolar confinement and the mirror-ratio lever

The central-cell ions are confined not by the mirror ratio alone but by the electrostatic potential peak the plug raises. With the plug electrons in Boltzmann equilibrium along the field line, the confining potential drop from central cell to plug is set by the density contrast,

$$\frac{e\Delta\varphi_c}{T_e} = \ln \frac{n_{\text{plug}}}{n_c}, \quad (1)$$

so the plug density $n_{\text{plug}} = 4 \times 10^{21} \text{ m}^{-3} \approx 16 n_c$ delivers $e\Delta\varphi_c/T_e \approx \ln 16 \approx 2.8$. A plug that cannot hold that density—because a microinstability degrades it—cannot raise the barrier, which is precisely why plug microstability (§3.2) is a confinement issue and not a cosmetic one. The population that leaks over the barrier and out the mirror throat is governed by the Pastukhov problem [13]: collisional scattering into the combined mirror-and-potential loss cone. For a species of temperature T confined by mirror ratio R and potential $e\varphi$, the collisional confinement time takes the Pastukhov–Cohen form [14]

$$\tau_{\parallel} = \tau_{ii} G(R) \frac{e\varphi}{T} \exp\left(\frac{e\varphi}{T}\right), \quad G(R) = \frac{(2R+1)}{2R} \ln(4R+2), \quad (2)$$

where τ_{ii} is the ion–ion self-collision time and $G(R)$ is the weak logarithmic mirror factor. The axial end-loss power scales as $P_{\text{ax}} \propto nV/\tau_{\parallel}$, so raising the mirror ratio $R \rightarrow R_{mc}$ lengthens $\tau_{\parallel}$ through $G(R_{mc})$ and suppresses P_{ax} . This is the lever of §3.4: mirror ratio buys confinement and hence gain, but at the cost of a higher plug field and a larger downstream expander.

3.2 The anisotropy-driven ion-cyclotron microinstability

The plug plasma is strongly mirror-confined and therefore anisotropic, $A = T_{\perp}/T_{\parallel} = 2$. Perpendicular free energy of this kind drives an electromagnetic ion-cyclotron (AIC / anisotropy) mode propagating along $\mathbf{B}_0$. Linearising the Vlasov–Maxwell system about a bi-Maxwellian hot-ion distribution

$$f_h(v_{\parallel}, v_{\perp}) = \frac{n_h}{\pi^{3/2} v_{t\parallel} v_{t\perp}^2} \exp\left(-\frac{v_{\parallel}^2}{v_{t\parallel}^2} - \frac{v_{\perp}^2}{v_{t\perp}^2}\right), \quad \frac{v_{t\perp}^2}{v_{t\parallel}^2} = A, \quad (3)$$

the parallel-propagating ion-cyclotron dispersion relation is

$$\frac{c^2 k_{\parallel}^2}{\omega^2} = 1 + \sum_s \frac{\omega_{ps}^2}{\omega^2} \left[(A_s - 1) + \frac{\omega - (1 - A_s)\omega_{ci}[s]}{k_{\parallel} v_{t\parallel,s}} Z(\zeta_s) \right], \quad \zeta_s = \frac{\omega - \omega_{ci}[s]}{k_{\parallel} v_{t\parallel,s}}, \quad (4)$$

with Z the plasma dispersion function, ω_{ps} the species plasma frequency and $\omega_{ci}[s]$ the species cyclotron frequency. The drive is the anisotropy factor $(A_s - 1)$; the mode is destabilised when the perpendicular free energy exceeds the magnetic and Landau restoring terms. A convenient reduced marginal-stability condition for the parallel firehose/anisotropy branch is

$$A - 1 \gtrsim \frac{1}{\beta_{\parallel}}, \quad \beta_{\parallel} = \frac{2\mu_0 n T_{\parallel}}{B_0^2}, \quad (5)$$

which at the plug point ($\beta_{\parallel} = 1.45$, from $n = 4 \times 10^{21} \text{ m}^{-3}$, $T_{\parallel} = 90 \text{ keV}$, $B_0 = 10 \text{ T}$) gives an instability margin $A - 1 = 1.0 > 1/\beta_{\parallel} = 0.69$: the un-stabilised plug is unstable, as the PIC confirms.

3.3 The kinetic-stabilizer mechanism

A kinetic stabilizer adds an isotropic energetic ion population at fraction f_s (isotropic: $A = 1$), leaving the hot anisotropic fraction $1 - f_s$. It quenches the mode by two routes that the reduced model makes explicit. First it lowers the aggregate anisotropy seen by the mode,

$$A_{\text{agg}}(f_s) = (1 - f_s) A + f_s \cdot 1 = A - f_s (A - 1) = 2 - f_s, \quad (6)$$

so that the marginal condition (5) predicts restabilisation when $A_{\text{agg}} - 1 \leq 1/\beta_{\parallel}$, i.e.

$$f_s^{\text{crit}} = A - 1 - \frac{1}{\beta_{\parallel}} = 1 - \frac{1}{1.45} \approx 0.31. \quad (7)$$

Second, and decisively, the stabilizer damps the mode resonantly only if its thermal speed matches the mode phase speed. For the isotropic 90 keV stabilizer the thermal speed is $v_{\text{warm}} = \sqrt{T_{\text{warm}}/m_i} = 1.86 \times 10^6 \text{ m s}^{-1} \approx 0.85 v_A$ (with $v_A = B_0/\sqrt{\mu_0 n m_i} = 2.18 \times 10^6 \text{ m s}^{-1}$), so the stabilizer is resonant with the ion-cyclotron mode. A cold 3 keV fill of the same fraction has $v_{\text{cold}} = 3.4 \times 10^5 \text{ m s}^{-1} \approx 0.16 v_A$ —far below resonance—and provides only density dilution, which (5) shows is insufficient. Equations (7) and the resonance estimate are the analytic content that the PIC of §5 tests directly, and the two agree: suppression sets in near $f_s \approx 0.30$.

3.4 Mirror ratio, engineering gain and the expander penalty

The engineering gain of the burner is the ratio of recovered electrical power to recirculated power. Writing P_{ch} for the charged-particle fusion power, f_{dir} for the direct-conversion fraction, P_{aux} for the auxiliary drive and $P_{\text{ax}}(R_{mc})$ for the mirror-ratio-dependent axial end loss that must be re-supplied, a reduced 0-D power balance gives

$$Q_E(R_{mc}) = \frac{f_{\text{dir}} P_{\text{ch}}}{P_{\text{inj}} + P_{\text{ax}}(R_{mc})}, \quad P_{\text{ax}}(R_{mc}) \propto \frac{1}{G(R_{mc}) \exp(e\varphi/T)}, \quad (8)$$

so that increasing R_{mc} suppresses P_{ax} through the Pastukhov factor (2) and raises Q_E . The penalty is geometric and magnetic: the plug field scales with R_{mc} at fixed central-cell field, $B_m = R_{mc} B_c$, and the end-loss stream—now more collimated—must be expanded over a proportionally larger area. The optimisation is therefore

$$\max_{R_{mc}} Q_E(R_{mc}) \quad \text{s.t.} \quad B_m(R_{mc}) \leq B_m^{\text{max}}, \quad A_{\text{exp}}(R_{mc}) \leq A_{\text{exp}}^{\text{max}}, \quad (9)$$

whose interior operating point is the design knee: the marginal gain dQ_E/dR_{mc} is nearly constant while the compounding downstream penalty (expander tank, conductor stress at the plug throat [4]) grows, so the burner is placed at the frozen $Q_E = 1.318$ rather than pushed to an extreme.

3.5 Expander flux conservation and the end-wall constraint

Whatever leaves the plug is nearly mono-directional, so the only lever on the end-wall load is geometric flux expansion. Conservation of magnetic flux along the diverging field tube, $B A = \text{const}$, means an expansion of the tube area by a factor $\Lambda = A_{\text{wall}}/A_{\text{thr}}$ drops the field, and

hence the parallel power density, by the same factor. With P_{ax} the axial power crossing a throat of area A_{thr} , the end-wall heat flux is

$$q_{\text{end}}(\Lambda) = \frac{P_{\text{ax}}}{A_{\text{thr}} \Lambda}, \quad (10)$$

an exact inverse law, $q_{\text{end}} \propto \Lambda^{-1}$. Imposing the survivable end-wall limit $q_{\text{end}} \leq q_{\text{lim}} = 2.5 \text{ MW m}^{-2}$ inverts to a hard floor on the expansion,

$$\Lambda_{\text{req}} = \frac{P_{\text{ax}}}{A_{\text{thr}} q_{\text{lim}}} = \frac{2.72 \times 10^3 \text{ MW}}{0.752 \text{ m}^2 \times 2.5 \text{ MW m}^{-2}} \approx 1447. \quad (11)$$

Below Λ_{req} the wall is over-limit; above it the wall is under-loaded at the cost of a larger expander tank. Equation (11) is the design floor, and it lands where the mirror-ratio choice (9) already put the expander area, closing the loop between the two studies.

4 Computational methodology: the NVIDIA GPU HPC campaign

All three levers were computed on the Kronos NVIDIA GPU HPC campaign, sharing a single verified numerical stack with the wider burner de-risking programme [1]. No cost or resource figures are reported; only the codes, the discretisations and the verification are described.

4.1 The 1D3V electromagnetic hybrid particle-in-cell model (AG-CX-03)

The kinetic stabilizer is evaluated with an electromagnetic hybrid PIC in the WarpX framework [29, 30], run in its 1D3V hybrid-Ohm configuration: kinetic ions on a massless, quasineutral, resistive electron fluid. The ion equations of motion are the relativistic-capable Boris push

$$\frac{d\mathbf{v}_p}{dt} = \frac{q_i}{m_i} (\mathbf{E} + \mathbf{v}_p \times \mathbf{B}), \quad \frac{d\mathbf{x}_p}{dt} = \mathbf{v}_p, \quad (12)$$

and the electric field is closed by the generalized Ohm's law for the electron fluid,

$$\mathbf{E} = -\mathbf{u}_i \times \mathbf{B} + \frac{1}{en} (\mathbf{J} \times \mathbf{B} - \nabla p_e) + \eta \mathbf{J} - \eta_H \nabla^2 \mathbf{J}, \quad (13)$$

with $\mathbf{J} = \mu_0^{-1} \nabla \times \mathbf{B}$, isothermal electrons ($T_e = 90 \text{ keV}$), resistivity $\eta = 10^{-6} \Omega \text{ m}$ and hyper-resistivity $\eta_H = 10^{-8} \Omega \text{ m}^3$; a density floor at $0.05 n_0$ and the Holmström vacuum treatment regularise low-density cells. Faraday's law $\partial_t \mathbf{B} = -\nabla \times \mathbf{E}$ is sub-cycled 40 times per ion step for numerical stability. The hot ions are loaded from the bi-Maxwellian (3) with $A = 2$; the stabilizer is an added isotropic 90 keV population at fraction f_s . Deuterons carry the campaign proxy mass $m_i = 2.5 m_p$.

Discretisation. The domain is a periodic line of $N_z = 1024$ cells at $\Delta z = \ell_i/10 = 5.69 \times 10^{-4} \text{ m}$ (ion skin depth $\ell_i = c/\omega_{pi}$), length $L_z = 0.583 \text{ m}$, with 1024 macroparticles per cell per species and a time step $\Delta t = 5 \times 10^{-3} t_{ci} = 8.20 \times 10^{-11} \text{ s}$ (cyclotron period $t_{ci} = 2\pi/\omega_{ci} = 16.4 \text{ ns}$, $\omega_{ci} = 3.83 \times 10^8 \text{ rad s}^{-1}$). Each case advances 6000 steps ($30 t_{ci}$). Four stabilizer fractions $f_s \in \{0, 0.15, 0.30, 0.45\}$ were run, one per GPU, sharing the build and numerics so the comparison is apples-to-apples.

Growth-rate diagnostic. The transverse magnetic-field energy $E_{\perp}(\tau) = \sum_z (B_x^2 + B_y^2)$ is sampled on a field-probe line every 8 steps. Because $E_{\perp} \propto e^{2\gamma t}$, the growth rate is extracted by sliding a fixed width- $4 t_{ci}$ window over the pre-saturation record, fitting $\ln E_{\perp}$ against $\tau = t/t_{ci}$, and selecting the steepest window with a clean exponential fit ($R^2 \geq 0.95$). The reported normalised rate is

$$\frac{\gamma}{\omega_{ci}} = \frac{1}{4\pi} \frac{d \ln E_{\perp}}{d\tau}, \quad (14)$$

the factor 4π following from $E_{\perp} \propto e^{2\gamma t}$ and $\omega_{ci} = 2\pi/t_{ci}$. The identical algorithm is applied to every fraction, so the relative trend is robust to the model-dependent absolute normalisation. A mandatory 250-step smoke test (finite growing transverse $\mathbf{B}$) preceded every production run.

4.2 Reduced-order mirror-ratio scan and field-line expander integration

The mirror-ratio optimum (AG-CX-06) is a 31-point reduced-order scan of the 0-D power balance (8) over $B_m = 11\text{--}26 \text{ T}$ (hence $R_{mc} = 2.98\text{--}7.07$ at fixed central-cell field), with the Pastukhov factor (2) supplying $P_{\text{ax}}(R_{mc})$ and the reduced power balance benchmarked against the WHAM operating point and the Pastukhov-plug design-space map of the companion study [5, 19]. The expander floor (AG-CX-11) is a 60-point field-line evaluation of the flux-conservation law (10)

over $\Lambda = 50\text{--}2500\times$, carrying the axial power $P_{\text{ax}} = 2.72 \times 10^3$ MW through a throat area $A_{\text{thr}} = 0.752$ m². Both reduced-order layers, the self-consistency checks and the orchestration run on the in-house HPC digital-twin node, so that the fastest analytic loops never contend with the PIC fleet, while the PIC and any downstream kinetic verification run on the NVIDIA GPU HPC campaign [32].

4.3 Verification, reproducibility and problem scale

Three levels of verification support the results. (i) The PIC growth diagnostic uses one algorithm across all fractions, so the f_s trend is invariant to the absolute-normalisation caveats of a reduced 1-D model; the baseline $f_s = 0$ rate ($\gamma/\omega_{ci} = 0.0365$, $R^2 = 0.99$) reproduces the plug-microstability gate of the wider campaign [6]. (ii) The expander law (10) is verified against its analytic inverse form: the 60 sampled points collapse onto the single line $q_{\text{end}} = (P_{\text{ax}}/A_{\text{thr}}) \Lambda^{-1}$ (figure 8), and the floor (11) is recovered to four significant figures ($1447.04\times$). (iii) The mirror-ratio scan holds the direct-conversion fraction f_{dir} within $\pm 3\%$ of 0.49 across the full range, confirming that the gain optimisation does not silently trade away the converter’s operating point. All three runs are register anchors—AG-CX-03, AG-CX-06 and AG-CX-11—in the Kronos Physics De-Risking Register [1], each with archived logs, and are regenerable from the deposited PIC inputs, reduced-order scan and field-line model. The PIC problem scale is $\sim 10^6$ macroparticles over 6000 time steps per case, four cases in parallel on the GPU fleet.

5 Results

5.1 Kinetic-stabilizer suppression

The energetic isotropic stabilizer monotonically quenches the plug microinstability (figure 2 a). The normalised growth rate falls

$$\frac{\gamma}{\omega_{ci}} : 0.0365 (f_s=0) \rightarrow 0.0239 (f_s=0.15) \rightarrow 0.0114 (f_s=0.30) \rightarrow \text{suppressed} (f_s=0.45), \quad (15)$$

a 35% reduction by $f_s = 0.15$, marginal by $f_s = 0.30$, and by $f_s = 0.45$ no clean exponential growth window survives. The saturated field-energy amplification collapses in step, from $145\times$ the initial level with no stabilizer to $50\times$, $23\times$ and finally $3\times$ (figure 2 b): the mode is effectively stabilised within the model. The suppression threshold near $f_s \approx 0.30$ matches the reduced marginal-stability estimate (7), $f_s^{\text{crit}} \approx 0.31$, so the kinetic result and the analytic condition agree. The design carries $f_s = 0.30$ as the selected minimum, with margin to 0.45.

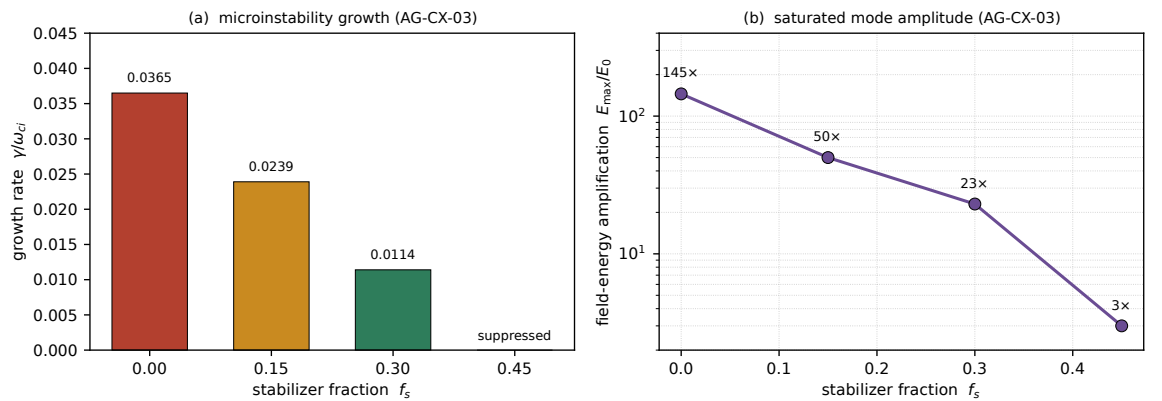


Figure 2. Kinetic-stabilizer end cell (AG-CX-03), from the 1D3V hybrid PIC. (a) Normalised growth rate γ/ω_{ci} versus stabilizer fraction f_s ; the mode is fully suppressed at $f_s = 0.45$. (b) Saturated transverse field-energy amplification E_{max}/E_0 (log scale) versus f_s , falling from $145\times$ to $3\times$. Both panels are the same four PIC runs, one per fraction, sharing the build and diagnostic.

The stabilising agent is the energetic, velocity-matched population, not density dilution. Figure 3 overlays the energetic (90 keV, AG-CX-03) stabilizer against a cold (3 keV, AG-CX-07) warm fill run in the same code with the same diagnostic. The cold fill leaves the growth rate flat ($\gamma/\omega_{ci} = 0.0366, 0.0354, 0.0335, 0.0393$ across $f = 0, 0.03, 0.0625, 0.10$)—it is inert—whereas the energetic population quenches the mode at comparable fractions. This is exactly the resonance argument of §3.3: the 90 keV stabilizer sits at $v_{\text{warm}} \approx 0.85 v_A$, resonant with the ion-cyclotron mode, while the 3 keV fill at $v_{\text{cold}} \approx 0.16 v_A$ is far off resonance and can only dilute.

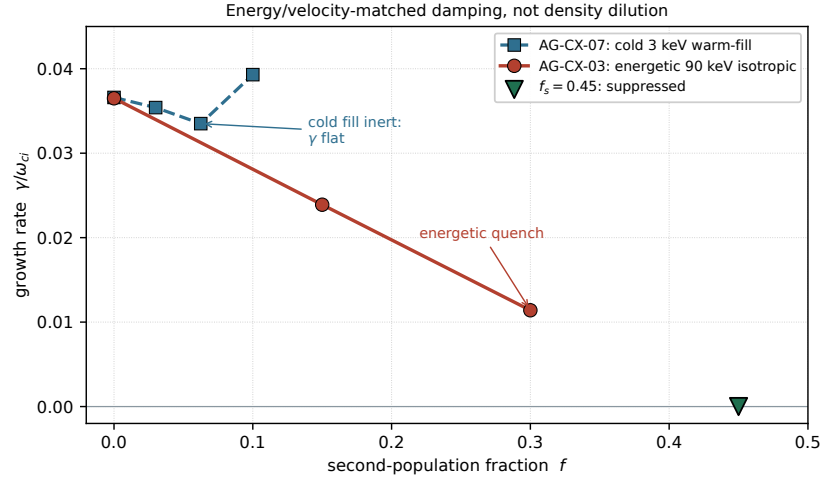


Figure 3. Energetic versus cold second population. The cold 3 keV warm fill (AG-CX-07, blue) leaves γ/ω_{ci} flat; the energetic 90 keV isotropic stabilizer (AG-CX-03, red) quenches the mode and is suppressed by $f_s = 0.45$. The contrast isolates energy/velocity-matched resonant damping as the mechanism.

5.2 The mirror-ratio optimum

Across the scan $R_{mc} = 2.98\text{--}7.07$ (on-axis mirror field 11–26 T) the engineering gain rises monotonically from $Q_E = 1.150$ to 1.555 (figure 4), a marginal gain $dQ_E/dR_{mc} \approx 0.099$ per unit mirror ratio that is nearly constant across the range. The required expander area grows in near-lockstep—by a relative factor $1017 \rightarrow 2095$, essentially linear in R_{mc} (figure 5)—while the direct-conversion fraction holds at $f_{dir} \approx 0.49$ throughout (within $\pm 3\%$). The gain therefore improves against a penalty (expander area, plug field, conductor stress) that grows without saturating, so the operating point is an interior knee, not an endpoint. The burner is fixed at

$$R_{mc} = 4.61, \quad B_m = 17 \text{ T}, \quad Q_E = 1.318, \quad (16)$$

the frozen burner engineering gain, behind a plug coil whose peak conductor field is 26.49 T—the same magnet’s governing structural number at this operating point, with thermal margin 37 K and load fraction 0.43 in the companion structural analysis [4, 1].

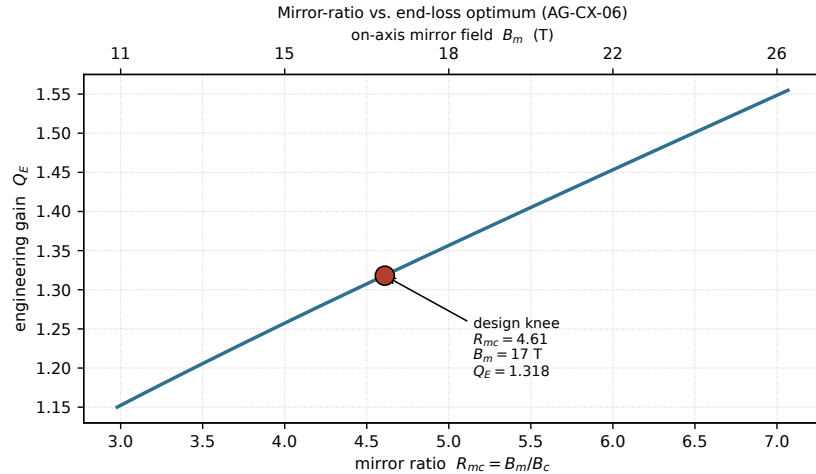


Figure 4. Mirror-ratio $\leftrightarrow$ end-loss optimum (AG-CX-06). Engineering gain Q_E versus central-cell mirror ratio R_{mc} (top axis: on-axis mirror field B_m). The design knee (red) sits at $R_{mc} = 4.61$, $B_m = 17 \text{ T}$, $Q_E = 1.318$ —the frozen burner point.

The trade is made quantitative by the gain-versus-penalty Pareto of figure 6: engineering gain plotted against the expander-area penalty, coloured by R_{mc} . Each increment of gain past the knee costs a fixed and growing slab of expander area and plug field, so the knee at $Q_E = 1.318$ is the point beyond which the marginal return no longer justifies the compounding downstream cost.

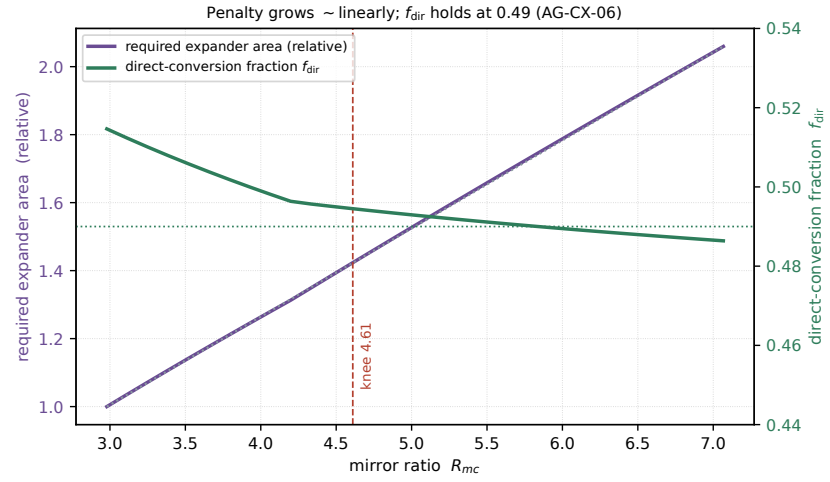


Figure 5. The mirror-ratio penalty (AG-CX-06). The required expander area (purple, relative) grows nearly linearly with R_{mc} —close to the linear reference (grey dotted)—while the direct-conversion fraction f_{dir} (green) holds at 0.49 across the scan. The design knee at $R_{mc} = 4.61$ is marked.

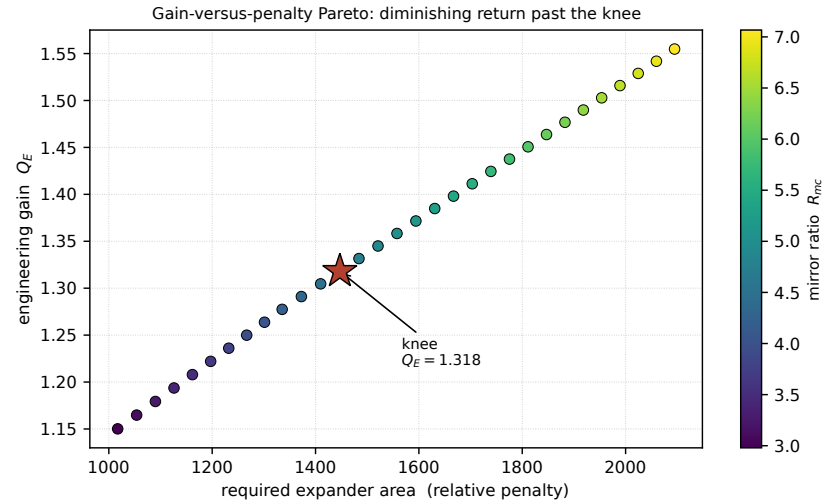


Figure 6. Gain-versus-penalty Pareto (AG-CX-06). Engineering gain Q_E against the relative expander-area penalty, coloured by mirror ratio R_{mc} ; the design knee ($Q_E = 1.318$) is the red star. Gain rises with diminishing return against a penalty that grows without saturating.

5.3 The expander floor

The end-wall heat flux obeys the inverse law (10) exactly. Scanning the flux expansion $\Lambda = 50\text{--}2500\times$ (figure 7), the flux falls from 36 MW m^{-2} at $\Lambda = 50$ through the 2.5 MW m^{-2} end-wall limit at

$$\Lambda_{\text{req}} = \frac{P_{\text{ax}}}{A_{\text{thr}} q_{\text{lim}}} = 1447\times, \quad (17)$$

the design floor. Below it the wall is over-limit; above it the wall is under-loaded at the expense of a larger expander tank. The inverse-law form is verified directly in figure 8: plotting q_{end} against Λ^{-1} collapses all 60 samples onto the single straight line of slope $P_{\text{ax}}/A_{\text{thr}}$, confirming $q_{\text{end}} \propto \Lambda^{-1}$ and recovering the floor to four significant figures. The expansion floor is set here for a 0.752 m^2 throat carrying the 2.72 GW axial end-loss power, and it is consistent with the expander area implied independently by the mirror-ratio knee (§5.2), closing the loop between the two studies.

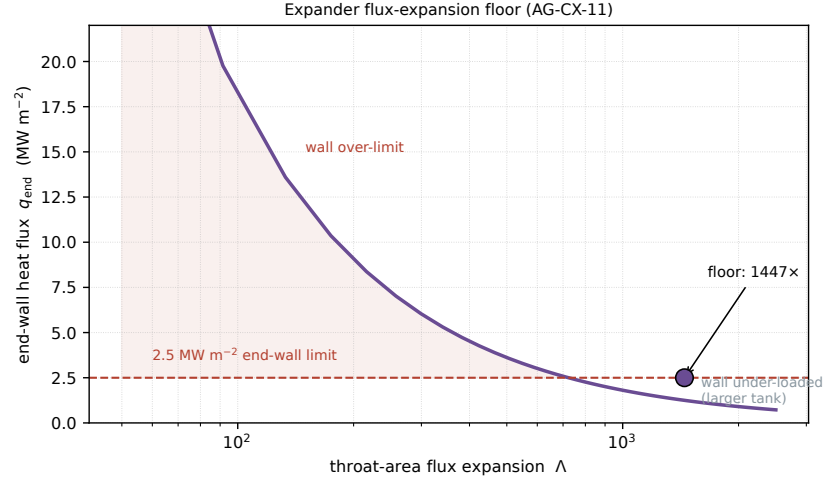


Figure 7. Expander flux-expansion floor (AG-CX-11). End-wall heat flux q_{end} versus throat-area flux expansion Λ (log axis); the 2.5 MW m^{-2} limit (red dashed) is reached at $\Lambda \approx 1447\times$ (purple), the design floor. Shading marks the over-limit region.

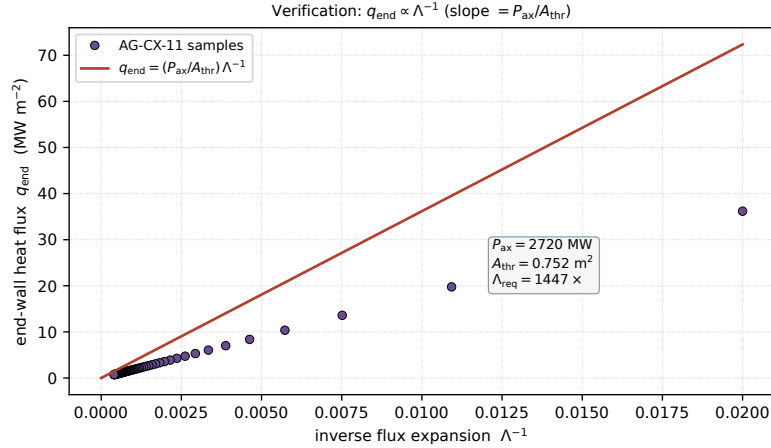


Figure 8. Verification of the inverse law (AG-CX-11). Plotting q_{end} against Λ^{-1} collapses the 60 field-line samples onto the single line $q_{\text{end}} = (P_{\text{ax}}/A_{\text{thr}}) \Lambda^{-1}$ (red), confirming $q_{\text{end}} \propto \Lambda^{-1}$. Inset values are the throat power and area that set the slope and floor.

5.4 The integrated end-cell design and two distinct magnets

The three levers define one coherent end cell (table 2). A kinetic stabilizer at $f_s \geq 0.30$ holds the plug micro-stable; a mirror ratio $R_{mc} = 4.61$ behind a plug coil of 26.49 T peak conductor field puts the burner at the gain knee $Q_E = 1.318$; and a $\approx 1447\times$ expander brings the end-loss stream to the 2.5 MW m^{-2} wall limit. The direct-conversion fraction of 0.49 that the downstream converter assumes [3] is preserved across the mirror-ratio choice, so the end-cell and

direct-energy-converter designs are mutually consistent. Every number is a tested point on its own study, not an assumption imported from another.

The magnets must not be conflated. Figure 9 separates the three field quantities that appear in this paper: the burner plug coil at 26.49 T peak conductor field, the burner on-axis mirror field 17 T at the knee, and the breeder centrepost at 16.84 T peak conductor field—two different magnets in two different machines, plus the plasma-region field of the burner. The plug coil is a discrete high-field solenoid engineered in the companion structural study [4]; the breeder centrepost is the spherical-tokamak breeder’s toroidal-field return, a wholly separate device [1].

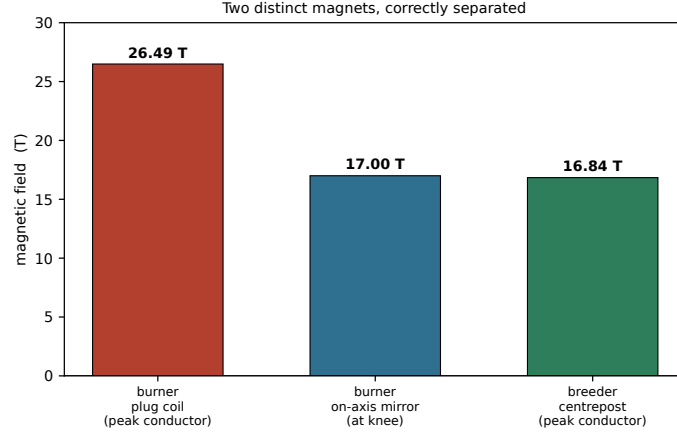


Figure 9. Two distinct magnets, correctly separated. The burner plug coil peak conductor field (26.49 T), the burner on-axis mirror field at the knee (17 T), and the breeder centrepost peak conductor field (16.84 T). The 26.49 T plug coil and the 16.84 T centrepost are different magnets in different machines.

Table 2. The integrated Aegis end-cell design point, with the governing equation and the de-risking serial for each quantity.

quantity	value	governing relation	serial
stabilizer fraction (min.) f_s	0.30	eq. (7), (14)	AG-CX-03
plug microinstability γ/ω_{ci}	0.0114 $\rightarrow$ suppressed	eq. (4)	AG-CX-03
mirror ratio R_{mc}	4.61	eq. (8), (9)	AG-CX-06
on-axis mirror field B_m	17 T	$B_m = R_{mc}B_c$	AG-CX-06
engineering gain Q_E	1.318 (frozen)	eq. (8)	AG-CX-06
direct-conversion fraction f_{dir}	0.49 (held)	eq. (8)	AG-CX-06
expander flux expansion Λ	$\approx 1447\times$	eq. (11)	AG-CX-11
end-wall heat flux q_{end}	$\leq 2.5 \text{ MW m}^{-2}$	eq. (10)	AG-CX-11
plug coil peak field	26.49 T	structural [4]	—

6 Discussion

6.1 Three coupled optima

The end-cell down-select is a study in coupled optima of three distinct mathematical types. Stability is a *threshold*: once f_s clears the marginal condition (7) near 0.30 the mode is gone, and there is no return on overspending the stabilizer beam. The mirror ratio is a genuine *trade with a knee*: gain rises with diminishing return (8) against a penalty (9) that grows without saturating, and the burner is placed on the trade at the frozen $Q_E = 1.318$ rather than at an extreme. The expander is a hard *floor* set by a wall limit (11), and it lands where the mirror-ratio choice already put the expander area. That the three studies reinforce one another—the stabilizer holds the density that raises the potential that the mirror ratio then exploits, and the expander area agrees between two independent derivations—is the signature of a consistent design rather than three separately tuned knobs.

6.2 Against the published benchmarks

The individual numbers sit comfortably in the mirror-fusion literature. The suppression of an anisotropy-driven ion-cyclotron mode by an energetic isotropic population is the microscopic analogue of the warm-stream stabilisation measured on 2XIIB, where a modest warm plasma quenched the loss-cone mode [23, 24]; the key quantitative distinction we draw—that the stabiliser

must be *energetic* (velocity-matched, $\approx 0.85 v_A$) rather than merely dense—is consistent with the resonance physics of the ion-cyclotron branch [16]. The marginal fraction $f_s \approx 0.30$ is a concrete, testable prediction that both the PIC (14) and the reduced threshold (7) return independently. The mirror-ratio optimum is anchored to the modern axisymmetric-mirror operating window: the on-axis mirror field of 17 T at $R_{mc} = 4.61$ is within the field envelope demonstrated by HTS axisymmetric plugs [19, 20], and the engineering-gain trend reproduces the design-space map benchmarked against WHAM in the companion study [5]. The kinetic-stabilizer route itself is the one advocated by Fowler, Moir and Simonen as the path to a simpler axisymmetric tandem mirror [17, 18]. The expander floor of $\approx 1450\times$ is a geometric consequence of the 2.5 MW m^{-2} end-wall limit and the 2.72 GW axial stream, well within the flux-expansion factors contemplated for mirror end tanks [15], and it feeds the direct converter whose recovery physics is treated separately [3, 25].

6.3 Why the plug potential depends on microstability

The through-line of the three studies is equation (1): the ambipolar barrier depends logarithmically on the plug density, so a micro-unstable plug that cannot hold $n_{\text{plug}} \approx 16 n_c$ cannot raise the potential the central cell needs. The kinetic stabilizer is therefore not an isolated fix but the enabler of the confinement that the mirror-ratio lever then optimises—and the sloshing-ion and thermal-barrier physics that shape the same potential are treated in the companion plug studies [7]. The three levers are one problem seen from three sides.

7 Limitations

The one honest gate is dimensionality. The stabilizer growth rates are from a 1D3V electromagnetic hybrid PIC, which resolves the parallel-propagating anisotropy microstability channel but not the full two-dimensional plug geometry, and in particular it cannot represent the good-curvature MHD interchange/ballooning role that a kinetic stabilizer also plays in a real tandem mirror. What is demonstrated here is narrower and stated as such: an added energetic isotropic population quenches the plug ion-cyclotron/anisotropy microinstability, with a threshold near $f_s \approx 0.30$ that the reduced marginal condition (7) independently predicts. A two-dimensional kinetic confirmation of the suppression—and of the MHD role—is the single open refinement; it is a strengthening of an already monotonic and complete 1-D result, not a reopening of it. The absolute growth-rate magnitudes are model-dependent lower bounds (finite resistivity, coarse Δz , single seed), while the relative f_s trend, computed with identical numerics across the four runs, is the robust deliverable.

8 Conclusion

The Aegis burner’s end cell is fixed by three tested levers that close on one design. A kinetic stabilizer at $f_s \geq 0.30$ suppresses the plug microinstability (growth $\gamma/\omega_{ci} = 0.0365 \rightarrow 0.0114 \rightarrow$ suppressed, amplitude $145\times \rightarrow 3\times$), and the suppression threshold agrees with the reduced marginal-stability estimate $f_s^{\text{crit}} \approx 0.31$; the stabilising mechanism is energy/velocity-matched resonant damping ($v_{\text{warm}} \approx 0.85 v_A$), not density dilution, as the contrast with a cold fill establishes. A mirror ratio $R_{mc} = 4.61$ —on-axis field 17 T behind a 26.49 T plug coil, distinct from the breeder’s 16.84 T centrepost—sits at the gain knee $Q_E = 1.318$, where gain rises with diminishing return against a nearly linear expander-area and plug-field penalty. And a $\approx 1447\times$ expander brings the 2.72 GW end-loss stream to the 2.5 MW m^{-2} wall limit through the exact inverse law $q_{\text{end}} \propto \Lambda^{-1}$. The three define one consistent end-cell design that delivers the frozen burner gain with the direct-conversion fraction held at 0.49, and a two-dimensional kinetic check of the stabilizer is the only refinement outstanding.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The particle-in-cell, reduced-order and field-line computations ran on the Kronos NVIDIA GPU HPC campaign alongside an in-house HPC node for the reduced-order and analytic layers. The authors thank the Kronos physics de-risking team for independent gate re-runs.

Reproducibility. The end-cell results are register anchors—AG-CX-03 (kinetic-stabilizer WarpX scan), AG-CX-06 (mirror-ratio optimum) and AG-CX-11 (expander flux-expansion)—in the Kronos Physics De-Risking Register [1], and are regenerable from the archived particle-in-cell inputs and logs, the reduced-order scan and the field-line model. This paper is archived at doi:10.5281/zenodo.22645939; official version:

<https://www.kronosfusionenergy.com/publications/tandem-mirror-end-cell-and-plug-alternatives-for-a-d-he.html>; its official version is at <https://www.kronosfusionenergy.com/publications/tandem-mirror-end-cell-and-plug-alternatives-for-a-d-he.html>; and the de-risking register is archived at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645939. The stabilizer growth-rate logs, mirror-ratio scan and expander sizing are archived with the deposit and reference the Kronos Physics De-Risking Register [1] doi:10.5281/zenodo.22133057 and the burner design paper [2]. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

The authors are affiliated with Kronos Fusion Energy. Provisional patent applications relating to aspects of this work have been filed.

References

- [1] P I Ford, *The Physics De-Risking Register: Gate-by-Gate Evidence Across a Compact Spherical-Tokamak Breeder and Its $D-^3\text{He}$ Tandem-Mirror Burners*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22133057; <https://www.kronosfusionenergy.com/publications/de-risking-register>
- [2] P I Ford and G L Kulcinski, *The Aegis Forward-Base Burner: Q_E Closure, Plug Requirement, and Mirror Confinement in an Islanded Low-Neutron $D-^3\text{He}$ Tandem Mirror*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645693; <https://www.kronosfusionenergy.com/publications/aegis-burner.html>
- [3] P I Ford, *Direct Energy Conversion for a $D-^3\text{He}$ Tandem-Mirror Burner: Recovering Charged-Particle Power Directly Without a Thermal Cycle*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645700; <https://www.kronosfusionenergy.com/publications/direct-energy-conversion.html>
- [4] P I Ford, R J Weggel and C Weggel, *Engineering the 26.49 T Plug Coil of a $D-^3\text{He}$ Tandem-Mirror Burner: Three-Dimensional Structural FEA, No-Insulation REBCO Quench Margin and Throat-Conductor Stress*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645915; <https://www.kronosfusionenergy.com/publications/engineering-the-26-49-t-plug-coil-of-a-d-he-tandem-mirr.html>
- [5] P I Ford, *Tandem-Mirror Design-Space Mapping for a $D-^3\text{He}$ Burner: 0-D Power Balance Benchmarked against WHAM and the Pastukhov-Plug Operating Point*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645911; <https://www.kronosfusionenergy.com/publications/tandem-mirror-design-space-mapping-for-a-d-he-burner-0-.html>
- [6] P I Ford, *Velocity-Space Microstability at the DCLC Knee: Warm-Fill Requirements and Residual AIC Activity in a $D-^3\text{He}$ Tandem-Mirror Burner*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645714; <https://www.kronosfusionenergy.com/publications/dclc-loss-cone-microstability.html>
- [7] P I Ford, *Sloshing-Ion Ambipolar-Potential Formation in a $D-^3\text{He}$ Tandem-Mirror Burner: The Density-Ratio-Potential Pareto and Non-Maxwellian Tail Control*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645913; <https://www.kronosfusionenergy.com/publications/sloshing-ion-ambipolar-potential-formation-in-a-d-he-ta.html>
- [8] P I Ford, *End-Cell, Expander, and Halo Physics in a $D-^3\text{He}$ Tandem-Mirror Burner: The Mirror Ends That Set the Loss Budget*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645852; <https://www.kronosfusionenergy.com/publications/end-cell-expander-and-halo-physics-in-a-d-3he-tandem-mi.html>
- [9] G I Dimov, V V Zakaidakov and M E Kishinevsky, Thermal-barrier / ambipolar (tandem) mirror trap, *Sov. J. Plasma Phys.* **2** 326 (1976).
- [10] T K Fowler and B G Logan, The tandem mirror reactor, *Comments Plasma Phys. Control. Fusion* **2** 167 (1977).
- [11] D E Baldwin and B G Logan, Improved tandem mirror fusion reactor, *Phys. Rev. Lett.* **43** 1318 (1979). doi:10.1103/PhysRevLett.43.1318
- [12] D E Baldwin, End-loss processes from a mirror machine, *Rev. Mod. Phys.* **49** 317 (1977).
- [13] V P Pastukhov, Collisional losses of electrons from an adiabatic trap in a plasma with a positive potential, *Nucl. Fusion* **14** 3 (1974). doi:10.1088/0029-5515/14/1/001
- [14] R H Cohen, M E Rensink, T A Cutler and A A Mirin, Collisional loss of electrostatically confined species in a magnetic mirror, *Nucl. Fusion* **18** 1229 (1978).
- [15] R F Post, The magnetic mirror approach to fusion, *Nucl. Fusion* **27** 1579 (1987). doi:10.1088/0029-5515/27/10/001
- [16] R F Post and M N Rosenbluth, Electrostatic instabilities in finite mirror-confined plasmas, *Phys. Fluids* **9** 730 (1966).
- [17] T K Fowler, R W Moir and T C Simonen, A new simpler way to obtain high fusion power gain in tandem mirrors, *Nucl. Fusion* **57** 056014 (2017). doi:10.1088/1741-4326/aa5e54
- [18] D D Ryutov, H L Berk, B I Cohen, A W Molvik and T C Simonen, Magneto-hydrodynamically stable axisymmetric mirrors, *Phys. Plasmas* **18** 092301 (2011).
- [19] D Endrizzi *et al*, Physics basis for the Wisconsin HTS Axisymmetric Mirror (WHAM), *J. Plasma Phys.* **89** 975890501 (2023). doi:10.1017/S0022377823000806
- [20] A A Ivanov and V V Prikhodko, Gas-dynamic trap: an overview of the concept and experimental results, *Plasma Phys. Control. Fusion* **55** 063001 (2013).
- [21] P A Bagryansky *et al*, Threefold increase of the bulk electron temperature of plasma discharges in a magnetic mirror device, *Phys. Rev. Lett.* **114** 205001 (2015).
- [22] M Inutake *et al*, Thermal barrier formation and plasma confinement in the axisymmetrized tandem mirror GAMMA-10, *Phys. Rev. Lett.* **55** 939 (1985). doi:10.1103/PhysRevLett.55.939
- [23] F H Coensgen *et al*, Stabilization of a neutral-beam-sustained, mirror-confined plasma, *Phys. Rev. Lett.* **44** 1132 (1980).

- [24] T A Casper and G R Smith, Anisotropy-driven ion-cyclotron instability in a mirror machine, *Phys. Rev. Lett.* **48** 1015 (1982).
- [25] R W Moir and W L Barr, Venetian-blind direct energy converter for fusion reactors, *Nucl. Fusion* **13** 35 (1973). doi:10.1088/0029-5515/13/1/005
- [26] W L Barr and R W Moir, Test results on plasma direct converters, *Nucl. Technol. Fusion* **3** 98 (1983).
- [27] W M Nevins, A review of confinement requirements for advanced fuels, *J. Fusion Energy* **17** 25 (1998).
- [28] G L Kulcinski and J F Santarius, Advanced fuels under debate, *Nature* **396** 724 (1998). doi:10.1038/25456
- [29] J-L Vay *et al*, Warp-X: a new exascale computing platform for beam–plasma simulations, *Nucl. Instrum. Methods Phys. Res. A* **909** 476 (2018). doi:10.1016/j.nima.2018.01.035
- [30] L Fedeli *et al*, Pushing the frontier in the design of laser-based electron accelerators with groundbreaking mesh-refined particle-in-cell simulations on exascale-class supercomputers, in *Proc. Int. Conf. for High Performance Computing (SC22)* (2022).
- [31] C K Birdsall and A B Langdon, *Plasma Physics via Computer Simulation* (McGraw-Hill, New York, 1985).
- [32] P Benner, S Gugercin and K Willcox, A survey of projection-based model reduction methods for parametric dynamical systems, *SIAM Rev.* **57** 483 (2015). doi:10.1137/130932715
- [33] J E Menard *et al*, Fusion nuclear science facilities and pilot plants based on the spherical tokamak, *Nucl. Fusion* **56** 106023 (2016). doi:10.1088/0029-5515/56/10/106023
- [34] B N Sorbom *et al*, ARC: a compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets, *Fusion Eng. Des.* **100** 378 (2015). doi:10.1016/j.fusengdes.2015.07.008
- [35] A J Creely *et al*, Overview of the SPARC tokamak, *J. Plasma Phys.* **86** 865860502 (2020). doi:10.1017/S0022377820001257
- [36] S Hahn *et al*, 45.5-tesla direct-current magnetic field generated with a high-temperature superconducting magnet, *Nature* **570** 496 (2019). doi:10.1038/s41586-019-1293-1